

OPERATIONS • ARTIFICIAL INTELLIGENCE • SMES

SmartFlow AI

A data-driven AI operations assistant for SMEs and producer cooperatives — customer communication, orders, shipments, stock, and daily tasks in a single workflow.

Pitch deck · 10 slides · July 2026 · Live demo: smartflow-frontend-production.up.railway.app

Problem

Small businesses typically run their operations across disconnected tools and manual tracking. This weakens the customer experience, stock visibility, and management response time.

Delayed responses

Messages, order status, and shipment status live in different places, so customer questions can't be answered quickly or consistently.

Late-detected risk

Critical stock shortages and shipment delays usually surface only after a customer complains.

Scattered tasks

Daily operational priorities depend on the person and the day, with no consolidated briefing.

Lost productivity

Searching for, copying, and reporting the same data over and over drains the team's capacity.

Solution: SmartFlow AI

A single AI agent that understands the customer's message, queries the business's **real operational data** (orders, products, shipments, stock) through structured tool calls, and produces **action-oriented** output for both the customer and the manager.

Core value proposition

- For customers: verifiable, data-backed answers (estimated delivery, delay days, stock availability).
- For managers: a summary dashboard, critical stock and shipment alerts, supplier email drafts, and a daily operations briefing.
- Not a scripted chatbot: a flow wired directly to the data layer via **function calling**.

Target audience & usage

Who is it for?

- Small and mid-sized producer / seller businesses.
- Producer cooperatives with a shared warehouse–shipping model.
- Teams handling high daily volume across the order + shipment + stock triangle.

User roles

- **Customer / end user:** natural-language questions about orders, products, and shipments.
- **Operations / manager:** dashboard, task list, alerts, and daily briefing.

MVP focus: a stable, deployed demo — single agent, structured data on SQLite; scaling and further integrations are the next phase.

Product features

Feature	Description
AI customer chat	POST /api/chat — intent and entity extraction; queries data by order number or product name.
Manager summary	GET /api/dashboard/summary — delayed shipments, critical stock, task and message overview.
Daily briefing	POST /api/tasks/generate — AI-generated operations tasks and priorities for the day.
Alert channel	POST /api/alerts/send — email alert to the manager (Resend integration).
Supplier communication	Supplier email draft generation for critical stock, shown directly in the dashboard.

Technical architecture

End-to-end flow: chat and dashboard clients → FastAPI → Gemini-powered agent with tool dispatch → SQLAlchemy / SQLite.

```
Customer Chat / Admin Dashboard
|
FastAPI
|
Gemini AI Agent + Tool Dispatcher
|
SQLAlchemy / SQLite
|
Orders, Products, Shipments, Tasks, Messages
```

When the Gemini API is unavailable, a **deterministic fallback** calls the same tool functions, so the demo and test suite keep working without an API key.

Covered scenarios (MVP)

#	Scenario	System coverage
1	Customer communication automation	Chat intents with data-backed replies
2	Order & product tracking	Order/product APIs and dashboard
3	Shipment process management	Delayed shipment detection and alert messages
4	Stock & inventory management	Critical stock threshold and supplier draft
5	Workflow & task management	Daily operations briefing and task summary

Out of scope for the MVP: advanced analytics/forecasting, a real WhatsApp API, live shipping-carrier APIs, a vector database, and multi-agent orchestration.

Technology stack

FastAPI

Python

SQLAlchemy 2

SQLite

Pydantic v2

Gemini API

Tool calling

Resend

React + Vite

Docker Compose

Railway (deployed)

pytest

AI approach

- Intent classes: order status, product info, shipment status, stock alert, daily briefing, general.
- Example tools: `get_order_status`, `get_product_info`, `get_cargo_status`, `check_stock_alerts`, `draft_supplier_email`, `generate_daily_briefing`, `send_manager_alert`.

The app's primary demo language is Turkish — the AI understands and replies in Turkish, and Turkish status strings (e.g. "Gecikmiş", "Kargoda") stay consistent with the business logic and test suite.

Roadmap

Today (MVP)

Single agent, SQLite, chat + dashboard + seed data; deployed live demo on Railway via Docker Compose; pytest contract test suite.

Short term

Real messaging channels (e.g. WhatsApp Business), and integration design for live shipping-carrier and payment providers.

Mid term

User authentication, rate limiting, hardened production deployment, observability.

Long term

Analytics and forecasting (Case 6), multi-warehouse support, and route-optimization modules.

Closing

"SmartFlow AI is not a scripted chatbot — it's an AI assistant that queries the business's operational data through tool / function calling and produces manager-ready actions."

SmartFlow AI — built for speed, transparency, and a predictable customer experience in SME operations.

Live demo: smartflow-frontend-production.up.railway.app · Source: README, architecture, and demo-script documentation · This PDF is generated from [docs/pitch-deck.html](#).